

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 14 — Hydrocarbons: Alkanes & Alkenes (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	14.1 Alkanes: Combustion, Free Radical Substitution, Catalytic Cracking · 14.2 Alkenes: Electrophilic Addition, Mo
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 14.1 Alkanes: Combustion, Substitution & Cracking** Complete vs incomplete combustion (CO toxicity, unburnt hydrocarbons, greenhouse warming); mechanism of
- 14.2 Alkenes: Electrophilic Addition & Oxidation** — Electrophilic addition of halogens, hydrogen halides, and steam (industrial hydration with H₃PO₄); Markovnikov's
- High-Frequency Core Repeats** — The 10 most frequently examined Cambridge Paper 1 questions on Hydrocarbon Chemistry (Q101 – Q110).

SUBTOPIC 14.1 — ALKANES: COMBUSTION, SUBSTITUTION & CRACKING (Q1 – Q40)

1 [EASY] Incomplete Combustion of Alkanes — 9701/11/M/J/23/Q17

[1]

What are the toxic primary products formed during the incomplete combustion of alkanes in a limited oxygen supply?

- [A] Carbon dioxide, CO₂, and sulfur dioxide, SO₂
- [B] Carbon monoxide, CO, and solid carbon particulate soot, C
- [C] Methane, CH₄, and ozone, O₃
- [D] Nitrogen dioxide, NO₂, and steam

2 [MEDIUM] Mechanism of Photochemical Chlorination of Alkanes — 9701/12/M/J/23/Q17

[1]

In the free radical chlorination of ethane under ultraviolet light, which equation represents an essential propagation step?

- [A] Cl₂ → 2Cl. (under UV irradiation)
- [B] CH₃CH₃ + Cl. → CH₃CH₂Cl + H.
- [C] CH₃CH₂. + Cl. → CH₃CH₂Cl
- [D] CH₃CH₂. + Cl₂ → CH₃CH₂Cl + Cl.

3 [EASY] Catalytic Cracking of Long-Chain Alkanes — 9701/13/M/J/23/Q17

[1]

What are the standard industrial conditions used for the catalytic cracking of long-chain hydrocarbons, and what types of products are formed?

- [A] Zeolite (aluminosilicate) catalyst at approximately 500 °C; produces shorter-chain branched alkanes and alkenes.
- [B] Concentrated sulfuric acid at 100 °C; produces carboxylic acids and esters.
- [C] Nickel catalyst at 150 °C; produces cycloalkanes and hydrogen gas.
- [D] Iron catalyst at 450 °C and 200 atm; produces aromatic compounds exclusively.

4 [HARD] Multi-Halogenation in Free Radical Substitution — 9701/11/O/N/23/Q17

[1]

When methane is reacted with an excess of chlorine gas in the presence of strong UV light, what is the ultimate principal organic product formed?

- [A] Chloromethane, CH₃Cl
- [B] Dichloromethane, CH₂Cl₂
- [C] Tetrachloromethane, CCl₄
- [D] Hexachloroethane, C₂Cl₆

5 [EASY] Homolytic Fission in Initiation — 9701/13/M/J/22/Q5

[1]

Why does the Cl-Cl bond break in preference to the C-H bond during the initiation stage of alkane chlorination?

- [A] The Cl-Cl bond enthalpy (242 kJ mol⁻¹) is much lower than the C-H bond enthalpy (413 kJ mol⁻¹), so UV photons selectively cleave Cl-Cl.
- [B] The C-H bond is completely ionic.
- [C] Chlorine molecules absorb infrared radiation rather than UV.
- [D] Carbon atoms repel ultraviolet radiation.

6 [HARD] Environmental Impact of Unburnt Hydrocarbons — 9701/11/M/J/22/Q6

[1]

What environmental problem is caused by unburnt hydrocarbons emitted from vehicle exhaust pipes?

- [A] They cause immediate ozone hole depletion in the stratosphere.
- [B] They acidify rain to pH 1.0.
- [C] They neutralize alkaline lake waters.
- [D] They react with nitrogen oxides and sunlight to form photochemical smog and ground-level ozone.

7 [EASY] Thermal vs Catalytic Cracking — 9701/12/M/J/222/Q7

[1]

How does thermal cracking differ from catalytic cracking in its primary product yield?

- [A] Thermal cracking requires platinum catalysts at room temperature.
- [B] Thermal cracking uses high temperature (700–1000 °C) and pressure to produce a high proportion of straight-chain alkenes (ethene, propene).
- [C] Thermal cracking only yields methane and hydrogen.
- [D] Thermal cracking produces exclusively cyclic aromatics.

8 [HARD] Mechanism of Bromine Abstraction — 9701/13/M/J/223/Q8

[1]

In the bromination of propane, why is 2-bromopropane the major mono-substituted product rather than 1-bromopropane?

- [A] Bromine atoms are too large to fit at the terminal position.
- [B] 1-bromopropane has a much higher boiling point.
- [C] The secondary 2-propyl radical formed as an intermediate is more stable than the primary 1-propyl radical due to the +I inductive effect of two methyl groups.
- [D] Secondary C–H bonds are completely unreactive.

9 [EASY] Combustion Stoichiometry of Propane — 9701/11/M/J/224/Q9

[1]

What volume of oxygen gas (measured at RTP) is required for the complete combustion of 1.0 dm³ of propane, C₃H₈?

- [A] 3.0 dm³
- [B] 5.0 dm³
- [C] 4.0 dm³
- [D] 10.0 dm³

10 [HARD] Greenhouse Effect and Alkanes — 9701/12/M/J/220/Q10

[1]

Why is methane (CH₄) a potent greenhouse gas compared to carbon dioxide?

- [A] Methane absorbs infrared radiation intensely across key atmospheric window wavelengths, having a global warming potential ~28 times greater than CO₂.
- [B] Methane reacts with water to form carbonic acid.
- [C] Methane destroys the Earth's magnetic field.
- [D] Methane forms a physical reflective cloud of ice in the mesosphere.

11 [EASY] Fractional Distillation of Crude Oil — 9701/13/M/J/221/Q11

[1]

On what physical property does the separation of crude oil fractions in a fractionating column depend?

- [A] Differences in chemical reactivity with water.
- [B] Differences in density of solid precipitates.
- [C] Differences in solubility in ethanol.
- [D] Differences in boiling points, which depend directly on molecular size and intermolecular London dispersion forces.

12 [HARD] Reactivity of Alkanes — 9701/11/M/J/222/Q12

[1]

Why are alkanes generally inert to attack by acids, bases, oxidizing agents, and nucleophiles at room temperature?

- [A] Alkanes possess delocalized pi electron clouds that shield the bonds.
- [B] Carbon atoms in alkanes are sp hybridised.
- [C] Alkanes contain only strong, non-polar C–C and C–H sigma bonds with no region of high or low electron density to attract attacking reagents.
- [D] Alkanes form strong hydrogen bonds that resist reaction.

13 [EASY] Catalytic Reforming of Alkanes — 9701/12/M/J/223/Q13

[1]

What is the purpose of catalytic reforming of straight-chain alkanes in petroleum refining?

- [A] To convert straight-chain alkanes into branched alkanes and cyclic/aromatic hydrocarbons to increase the octane rating of fuels.
- [B] To convert alkanes into synthetic detergents.
- [C] To completely crack all fuel into pure hydrogen.
- [D] To oxidize alkanes into carboxylic acids.

14 [HARD] Combustion Analysis of Pentane — 9701/13/M/J/224/Q14

[1]

How many moles of water are produced by the complete combustion of one mole of pentane, C_5H_{12} ?

- [A] 5 moles
- [B] 12 moles
- [C] 6 moles
- [D] 10 moles

15 [EASY] Propagation Steps in Halogenation — 9701/11/M/J/220/Q15

[1]

What is the net energetic characteristic of the two propagation steps in alkane chlorination?

- [A] Propagation requires constant input of external UV light.
- [B] The overall propagation sequence is exothermic, providing the thermodynamic driving force for continuous chain reaction.
- [C] Propagation creates two radicals from one radical.
- [D] Propagation forms ionic salts.

16 [HARD] Limiting Multi-Substitution — 9701/12/M/J/221/Q16

[1]

How can the reaction conditions be adjusted to maximize the yield of monochloromethane (CH_3Cl) over higher chlorinated products?

- [A] Use a large excess of chlorine gas.
- [B] Carry out the reaction at 500 °C in darkness.
- [C] Add concentrated sulfuric acid as a catalyst.
- [D] Use a large excess of methane relative to chlorine.

17 [EASY] Alkanes as Non-Polar Solvents — 9701/13/M/J/222/Q17

[1]

Why are liquid alkanes (such as hexane) immiscible with water but excellent solvents for non-polar grease?

- [A] Hexane reacts vigorously with water to form hexanol.
- [B] Water molecules are non-polar and repel hexane.
- [C] Hexane molecules cannot form hydrogen bonds with water molecules; the London forces between hexane and grease molecules are favorable for dissolution.
- [D] Hexane has a very high dielectric constant.

18 [HARD] Isomer Distribution in Free Radical Chlorination — 9701/11/M/J/223/Q18

[1]

When propane reacts with chlorine in UV light, what are the two mono-chloro structural isomers formed?

- [A] 1-chloropropane and 2-chloropropane
- [B] 1-chloropropane and 3-chloropropane
- [C] 2-chloropropane and 2,2-dichloropropane
- [D] Chlorocyclopropane and chloropropene

19 [EASY] Termination Step Combining Alkyl Radicals — 9701/12/M/J/224/Q19

[1]

In the photochemical bromination of bromoethane, which species would NOT be formed in a termination step?

- [A] 1,2-dibromoethane
- [B] 1,1-dibromoethane
- [C] 1,4-dibromobutane
- [D] Hydrogen gas, H₂

20 [HARD] Relative Reactivity: Fluorine vs Chlorine vs Bromine vs Iodine — 9701/13/M/J/220/Q20

[1]

Which halogens react explosively and which does not react at all with alkanes under standard photochemical conditions?

- [A] Chlorine is inert; iodine is explosive.
- [B] Fluorine reacts explosively; iodine does not react at all (endothermic).
- [C] Bromine is explosive; fluorine is inert.
- [D] All halogens react at identical rates.

21 [EASY] Octane Rating and Branching — 9701/11/M/J/221/Q21

[1]

Why do highly branched alkanes burn more smoothly in petrol engines than straight-chain isomers?

- [A] Branched alkanes have higher boiling points than straight chains.
- [B] Branching slows the formation of pre-ignition peroxides during compression, preventing engine 'knocking'.
- [C] Branched alkanes have higher carbon-to-hydrogen ratios.
- [D] Branched alkanes are completely non-flammable.

22 [HARD] Carbon Footprint of Fuels — 9701/12/M/J/222/Q22

[1]

Which alkane releases the LEAST mass of CO₂ per kilojoule of energy released upon complete combustion?

- [A] Propane, C₃H₈
- [B] Octane, C₈H₁₈
- [C] Hexadecane, C₁₆H₃₄
- [D] Methane, CH₄

23 [EASY] Chain Length and Viscosity — 9701/13/M/J/223/Q23

[1]

How does the viscosity of liquid alkanes change as carbon chain length increases?

- [A] Viscosity increases because longer chains have greater surface contact area and stronger London dispersion forces that tangle molecules.
- [B] Viscosity decreases because longer chains flow faster.
- [C] Viscosity remains completely constant.
- [D] Viscosity increases due to formation of covalent crosslinks.

24 [HARD] Free Radical Inhibitors — 9701/11/M/J/224/Q24

[1]

Why does the addition of trace amounts of oxygen gas temporarily retard the photochemical chlorination of methane?

- [A] Oxygen dissolves all chlorine gas.
- [B] Oxygen cools the reaction vessel to -50 °C.
- [C] Oxygen reacts with alkyl free radicals to form unreactive peroxy radicals (ROO $\cdot$), terminating chain propagation.
- [D] Oxygen absorbs ultraviolet light completely.

25 [EASY] Cracking of Dodecane — 9701/12/M/J/220/Q25**[1]**

In the cracking of dodecane ($\text{C}_{12}\text{H}_{26}$), one molecule yields one molecule of octane (C_8H_{18}) and two molecules of an identical alkene. What is the alkene?

- [A] Ethene, C_2H_4
- [B] Propene, C_3H_6
- [C] But-1-ene, C_4H_8
- [D] Methane, CH_4

26 [HARD] Combustion of Cycloalkanes — 9701/13/M/J/221/Q26**[1]**

What is the balanced equation for the complete combustion of one mole of cyclohexane (C_6H_{12})?

- [A] $\text{C}_6\text{H}_{12} + 6\text{O}_2 \rightarrow 6\text{CO} + 6\text{H}_2\text{O}$
- [B] $\text{C}_6\text{H}_{12} + 12\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2$
- [C] $\text{C}_6\text{H}_{12} + 8\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$
- [D] $\text{C}_6\text{H}_{12} + 9\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$

27 [EASY] Pollutants in Exhaust Gases — 9701/11/M/J/222/Q27**[1]**

Which compound is NOT a direct pollutant emitted in internal combustion engine exhaust gases?

- [A] Carbon monoxide, CO
- [B] Ozone, O_3
- [C] Nitrogen monoxide, NO
- [D] Unburnt hydrocarbons

28 [HARD] Stability of Alkyl Radicals — 9701/12/M/J/223/Q28**[1]**

Why is the tertiary butyl radical, $(\text{CH}_3)_3\text{C}\cdot$, significantly more stable than the primary butyl radical?

- [A] Tertiary radicals have complete electron octets.
- [B] Tertiary radicals are planar while primary radicals are spherical.
- [C] Hyperconjugation and the electron-donating inductive effect (+I) of three methyl groups delocalise the unpaired electron.
- [D] Tertiary radicals form hydrogen bonds.

29 [EASY] Boiling Point of Alkane Isomers — 9701/13/M/J/224/Q29**[1]**

Why does 2,2-dimethylpropane have a lower boiling point (9.5°C) than pentane (36.1°C) despite having the identical molecular mass?

- [A] Pentane contains polar covalent bonds.
- [B] Its spherical, branched shape provides less surface contact area between molecules, resulting in weaker London dispersion forces.
- [C] 2,2-dimethylpropane forms intramolecular hydrogen bonds.
- [D] Pentane has a higher vapor pressure.

30 [HARD] Homologous Series Enthalpy of Combustion — 9701/11/M/J/220/Q30**[1]**

How does the standard enthalpy of combustion (ΔH_c) change per additional $-\text{CH}_2-$ unit down the alkane homologous series?

- [A] It becomes more exothermic by approximately -650 kJ mol^{-1} per $-\text{CH}_2-$ group.
- [B] It becomes endothermic.
- [C] It decreases to zero.
- [D] It doubles for every carbon atom.

31 [EASY] Free Radical Dimerisation — 9701/12/M/J/221/Q31

[1]

During the photobromination of 2-methylpropane, what is the hydrocarbon formed if two tertiary butyl radicals undergo termination?

- [A] 2,2,4-trimethylpentane
- [B] Octane
- [C] 2,3-dimethylbutane
- [D] 2,2,3,3-tetramethylbutane

32 [HARD] Soot Particle Health Hazards — 9701/13/M/J/222/Q32

[1]

What is the primary human health hazard associated with PM_{2.5} carbon soot particles from diesel engines?

- [A] They cause immediate tooth decay.
- [B] They neutralize stomach hydrochloric acid.
- [C] They penetrate deep into the alveoli of the lungs, carrying carcinogenic polycyclic aromatic hydrocarbons into the bloodstream.
- [D] They deplete calcium from bones.

33 [EASY] Oxidation of Alkanes by KMnO₄ — 9701/11/M/J/223/Q33

[1]

What is observed when an alkane (such as hexane) is shaken with cold, acidified potassium manganate(VII)?

- [A] No reaction occurs; the purple color of KMnO₄ remains unchanged.
- [B] The purple solution turns green immediately.
- [C] A brown precipitate of MnO₂ forms with vigorous effervescence.
- [D] The hexane is oxidized to hexanoic acid.

34 [HARD] Industrial Source of Alkanes — 9701/12/M/J/224/Q34

[1]

What is the primary natural source from which alkanes are industrially extracted?

- [A] Fermentation of sugar cane
- [B] Electrolysis of brine
- [C] Crude petroleum oil and natural gas
- [D] Destructive distillation of wood

35 [EASY] Radical Abstraction Selectivity — 9701/13/M/J/220/Q35

[1]

Why is bromine significantly more regioselective than chlorine in free radical substitution reactions of alkanes?

- [A] Bromine is a smaller atom than chlorine.
- [B] The hydrogen abstraction step by Br. is endothermic and has a late transition state resembling the carbocation/radical intermediate, whereas for Cl. it is exothermic with an early transition state.
- [C] Bromine radicals are positively charged.
- [D] Chlorine cannot form secondary radicals.

36 [HARD] Catalytic Converter Chemistry for Alkanes — 9701/11/M/J/221/Q36

[1]

What reaction takes place on the platinum/palladium catalyst in a three-way catalytic converter to remove unburnt alkanes?

- [A] Reduction to solid carbon and hydrogen gas.
- [B] Hydrolysis by exhaust steam to alcohols.
- [C] Polymerisation into solid plastics.
- [D] Complete oxidation by excess oxygen: $C_xH_y + (x + y/4)O_2 \rightarrow xCO_2 + y/2 H_2O$

37 [EASY] Methane Hydrates — 9701/12/M/J/222/Q37**[1]**

What are methane hydrates (clathrates) found in deep ocean sediments and permafrost?

- [A] Solid salts of methane and sodium hydroxide.
- [B] Liquid solutions of methane in liquid ammonia.
- [C] Crystalline water ice cages that trap methane gas molecules under high pressure and low temperature.
- [D] Polymeric chains of methane linked by covalent bonds.

38 [HARD] Thermodynamic Stability of Alkanes — 9701/13/M/J/223/Q38**[1]**

Although alkanes are thermodynamically unstable with respect to combustion in air (exothermic), why are they kinetically stable at room temperature?

- [A] They possess a very high activation energy barrier due to the need to break strong C-H and C-C covalent bonds.
- [B] Their combustion is an endothermic equilibrium.
- [C] They are protected by an impenetrable electrostatic shield.
- [D] Oxygen molecules cannot collide with alkanes at 25 °C.

39 [EASY] Free Radical Chlorination in Darkness — 9701/11/M/J/224/Q39**[1]**

What happens when a mixture of methane and chlorine gas is kept at room temperature in complete darkness?

- [A] The mixture explodes violently within 5 seconds.
- [B] Methane is converted slowly into carbon soot.
- [C] Chlorine gas condenses into a yellow liquid.
- [D] No reaction occurs because no UV photons are present to supply the activation energy for homolytic fission of Cl₂.

40 [HARD] Cracking Byproduct Uses — 9701/12/M/J/220/Q40**[1]**

Why are the alkene byproducts of catalytic cracking (such as ethene and propene) of such immense economic importance to the chemical industry?

- [A] They are used directly as aircraft lubricating oils.
- [B] They serve as reactive raw monomers for the manufacture of addition polymers (polyethene, polypropene) and alcohols.
- [C] They are used as non-toxic fire extinguishants.
- [D] They are used to sweeten fruit juices.

SUBTOPIC 14.2 — ALKENES: ELECTROPHILIC ADDITION & OXIDATION (Q41 – Q100)**41** [EASY] Mechanism of Bromine Addition to Ethene — 9701/11/M/J/23/Q18**[1]**

What type of reaction mechanism occurs when ethene reacts with bromine dissolved in an inert organic solvent, and what is the visual observation?

- [A] Nucleophilic substitution; a white precipitate is formed.
- [B] Electrophilic addition; the orange-brown bromine solution is immediately decolorized.
- [C] Free radical substitution; the solution turns intense purple.
- [D] Elimination; bubbles of hydrogen bromide gas are evolved.

42 [HARD] Markovnikov Addition of HBr to Propene — 9701/12/M/J/23/Q18**[1]**

When hydrogen bromide, HBr, reacts with propene (CH₃-CH=CH₂), what is the major organic product and what explains its formation?

- [A] 1-bromopropane; the primary carbocation is more stable due to steric factors.
- [B] 1,2-dibromopropane; both carbons add bromine simultaneously.
- [C] Cyclopropane; HBr causes cyclization of the propene chain.
- [D] 2-bromopropane; the reaction proceeds via the more stable secondary carbocation intermediate (CH₃-CH⁺-CH₃).

43 [EASY] Oxidation of Alkenes by Cold, Dilute KMnO_4 — 9701/13/M/J/23/Q18**[1]**

What organic product and color change occur when propene is shaken with cold, dilute, acidified potassium manganate(VII)?

- [A] Propane-1,2-diol is formed; the purple solution is decolorized (or forms a brown precipitate of MnO_2 in neutral conditions).
- [B] Carbon dioxide and ethanoic acid are formed with vigorous effervescence.
- [C] Propanone is formed with no color change.
- [D] Propanoic acid is formed; the solution turns bright green.

44 [HARD] Oxidative Cleavage of Alkenes by Hot, Concentrated KMnO_4 — 9701/11/O/N/23/Q18**[1]**

An alkene X is heated under reflux with hot, concentrated, acidified potassium manganate(VII). The only organic products obtained are propanone (CH_3COCH_3) and ethanoic acid (CH_3COOH).

What is the structural formula of alkene X?

- [A] 2-methylbut-1-ene, $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}_2\text{CH}_3$
- [B] Pent-2-ene, $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2\text{CH}_3$
- [C] 2-methylbut-2-ene, $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_3$
- [D] 3-methylbut-1-ene, $\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)_2$

45 [EASY] Industrial Hydration of Ethene — 9701/11/M/J/220/Q45**[1]**

What catalyst and conditions are used in the direct industrial manufacture of ethanol from ethene and steam?

- [A] Concentrated phosphoric(V) acid (H_3PO_4) adsorbed on silica at 300°C and 60 atm pressure.
- [B] Nickel catalyst at 150°C and 1 atm.
- [C] Concentrated sulfuric acid at 0°C .
- [D] Iron catalyst at 450°C and 200 atm.

46 [HARD] Catalytic Hydrogenation of Alkenes — 9701/12/M/J/221/Q46**[1]**

What catalyst, temperature, and industrial application are associated with the addition of hydrogen to alkenes?

- [A] Iron catalyst at 450°C ; used in cracking petroleum.
- [B] Vanadium(V) oxide at 450°C ; used in making sulfuric acid.
- [C] Zeolite catalyst at 500°C ; used in making branched fuels.
- [D] Finely divided Nickel catalyst at 150°C (or Pt at RTP); used in hardening vegetable oils to make margarine.

47 [EASY] Electrophilic Addition of Bromine Water — 9701/13/M/J/222/Q47**[1]**

When propene is bubbled through aqueous bromine water ($\text{Br}_2(\text{aq})$), what is the major organic product formed?

- [A] 1,2-dibromopropane
- [B] 1-bromopropan-2-ol, $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2\text{Br}$
- [C] 2-bromopropan-1-ol
- [D] Propan-2-ol

48 [HARD] Cleavage Products of Terminal $=\text{CH}_2$ Group — 9701/11/M/J/223/Q48**[1]**

When a terminal alkene containing a $=\text{CH}_2$ group is treated with hot, concentrated, acidified KMnO_4 , what oxidation products are formed from that terminal group?

- [A] Methanoic acid (HCOOH)
- [B] Methanal (HCHO)
- [C] Carbon dioxide (CO_2) and water (H_2O)
- [D] Methanol (CH_3OH)

49 [EASY] Cleavage Products of 2,3-dimethylbut-2-ene — 9701/12/M/J/224/Q49**[1]**

What single organic product is formed when 2,3-dimethylbut-2-ene, $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$, is heated with hot, concentrated, acidified KMnO_4 ?

- [A] Ethanoic acid
- [B] Propanone, CH_3COCH_3
- [C] 2-methylpropanoic acid
- [D] Carbon dioxide

50 [HARD] Addition of Chlorine to But-2-ene — 9701/13/M/J/220/Q50**[1]**

What is the systematic name of the product formed by the addition of chlorine to but-2-ene?

- [A] 2,3-dichlorobutane
- [B] 1,2-dichlorobutane
- [C] 1,4-dichlorobutane
- [D] 2-chlorobutane

51 [EASY] Carbocation Rearrangement Likelihood — 9701/11/M/J/221/Q51**[1]**

Why does 3,3-dimethylbut-1-ene yield 2-bromo-2,3-dimethylbutane as the predominant product upon addition of HBr ?

- [A] Bromine directly displaces a methyl group.
- [B] Hydrogen attacks both ends of the double bond.
- [C] The reaction proceeds via a free radical mechanism.
- [D] The initial secondary carbocation undergoes a 1,2-methyl shift to form a more stable tertiary carbocation before attack by bromide.

52 [HARD] Test for Unsaturation — 9701/12/M/J/222/Q52**[1]**

Which observation confirms that an unknown liquid hydrocarbon contains an alkene $\text{C}=\text{C}$ double bond?

- [A] Effervescence with sodium carbonate solution.
- [B] A silver mirror with Tollens' reagent.
- [C] Decolorization of bromine water from orange-brown to colorless without needing UV light.
- [D] A brick-red precipitate with Fehling's solution.

53 [EASY] Stability Order of Carbocations — 9701/13/M/J/223/Q53**[1]**

Why is $(\text{CH}_3)_3\text{C}^+$ more stable than $\text{CH}_3\text{CH}_2\text{CH}_2^+$?

- [A] Three electron-releasing methyl groups disperse the positive charge across the molecule via the +I inductive effect, lowering electrostatic potential.
- [B] The primary carbocation is too small to exist.
- [C] The tertiary carbocation has a complete outer octet of 10 electrons.
- [D] The primary carbocation forms hydrogen bonds.

54 [HARD] Addition of Interhalogens (ICl) — 9701/11/M/J/224/Q54**[1]**

When iodine monochloride (I-Cl) reacts with propene, which atom acts as the electrophile and what is the major product?

- [A] Chlorine acts as the electrophile; major product is 1-chloro-2-iodopropane.
- [B] Both atoms act as nucleophiles.
- [C] Iodine acts as the electrophile ($\text{I}^{\delta+}$) because chlorine is more electronegative; major product is 2-chloro-1-iodopropane.
- [D] Iodine replaces a hydrogen on the methyl group.

55 [EASY] Deducing Alkene from KMnO_4 Oxidation — 9701/12/M/J/220/Q55**[1]**

Heating an alkene with hot concentrated KMnO_4 yields only butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$. What is the identity of the alkene?

- [A] Oct-1-ene
- [B] Oct-4-ene, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_2\text{CH}_3$
- [C] Hex-3-ene
- [D] But-1-ene

56 [HARD] Addition Polymerisation of Alkenes — 9701/13/M/J/221/Q56**[1]**

What type of reaction converts ethene molecules into polyethene, and what change in bonding occurs?

- [A] Condensation polymerisation; water is eliminated.
- [B] Elimination; hydrogen gas is evolved.
- [C] Substitution; chlorine atoms replace hydrogens.
- [D] Addition polymerisation; pi bonds break and new C-C sigma bonds form between monomer units.

57 [EASY] Stereochemistry of Alkene Addition — 9701/11/M/J/222/Q57**[1]**

Why does the bromination of cyclohexene yield trans-1,2-dibromocyclohexane rather than the cis isomer?

- [A] The cis isomer is too reactive to isolate.
- [B] Bromine molecules can only attack from the top face.
- [C] The reaction proceeds via a cyclic bromonium ion intermediate, forcing the incoming bromide ion to attack from the opposite face (anti-addition).
- [D] The reaction proceeds via free radical substitution.

58 [HARD] Addition of Sulfuric Acid to Alkenes — 9701/12/M/J/223/Q58**[1]**

What is the product when ethene is absorbed into concentrated sulfuric acid at room temperature, and what happens upon subsequent boiling with water?

- [A] Ethyl hydrogensulfate is formed; boiling with water hydrolyzes it to ethanol.
- [B] Ethanoic acid is formed directly.
- [C] Diethyl ether is formed as a gas.
- [D] Ethene is oxidized to carbon dioxide.

59 [EASY] Reactivity: Alkenes vs Benzene — 9701/13/M/J/224/Q59**[1]**

Why do alkenes readily undergo electrophilic addition while benzene undergoes electrophilic substitution?

- [A] Benzene contains only single sigma bonds.
- [B] Alkenes have lower electron density than benzene.
- [C] Benzene is non-planar.
- [D] Addition to benzene would destroy its exceptionally stable delocalised 6 pi electron aromatic resonance system.

60 [HARD] Oxidation of Cyclohexene — 9701/11/M/J/220/Q60**[1]**

What is the product when cyclohexene is heated with hot, concentrated, acidified KMnO_4 ?

- [A] Cyclohexanol
- [B] Hexanedioic acid (adipic acid), $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$
- [C] Cyclohexanone
- [D] Carbon dioxide and water only

61 [EASY] Heterolytic Cleavage of Halogen Molecule — 9701/12/M/J/221/Q61**[1]**

During electrophilic addition of Br₂, what causes the non-polar bromine molecule to become polarized?

- [A] The solvent permanently ionizes bromine molecules.
- [B] The high electron density in the C=C pi bond repels the electron cloud of the approaching Br₂ molecule, inducing a temporary dipole (Br δ⁺–Br δ[–]).
- [C] Ultraviolet light creates ionic dipoles.
- [D] Air pressure squeezes the bromine electrons.

62 [HARD] Markownikoff Addition with Water — 9701/13/M/J/222/Q62**[1]**

What is the major product when 2-methylpropene is reacted with steam in the presence of an acid catalyst?

- [A] 2-methylpropan-1-ol (primary alcohol)
- [B] Butan-2-ol
- [C] Butan-1-ol
- [D] 2-methylpropan-2-ol (tertiary alcohol)

63 [EASY] Oxidation of 2-methylbut-1-ene — 9701/11/M/J/223/Q63**[1]**

What products are obtained when 2-methylbut-1-ene, CH₂=C(CH₃)CH₂CH₃, is oxidized with hot concentrated acidified KMnO₄?

- [A] Carbon dioxide (CO₂), water (H₂O), and butanone (CH₃COCH₂CH₃)
- [B] Ethanoic acid and propanoic acid
- [C] 2-methylbutanoic acid only
- [D] Methanoic acid and butan-2-ol

64 [HARD] Addition of HCl to Alkenes vs HBr — 9701/12/M/J/224/Q64**[1]**

Why does electrophilic addition of HCl to alkenes occur more slowly than addition of HBr under identical conditions?

- [A] Chlorine is less electronegative than bromine.
- [B] Chloride ions are too large to attack carbocations.
- [C] The H–Cl bond enthalpy (431 kJ mol^{–1}) is higher than that of H–Br (366 kJ mol^{–1}), resulting in a higher activation energy for electrophilic protonation.
- [D] HCl exists only as a solid at room temperature.

65 [EASY] Identification of Alkene Structure from Gas Volume — 9701/13/M/J/220/Q65**[1]**

Complete combustion of 20 cm³ of a gaseous alkene requires 90 cm³ of oxygen gas at RTP. What is the alkene?

- [A] Propene, C₃H₆
- [B] Ethene, C₂H₄
- [C] But-1-ene, C₄H₈
- [D] Pent-1-ene, C₅H₁₀

66 [HARD] Relative Rates of Halogen Addition — 9701/11/M/J/221/Q66**[1]**

Which halogen reacts most rapidly with ethene in electrophilic addition?

- [A] Iodine > Bromine > Chlorine
- [B] All halogens react at identical rates
- [C] Fluorine cannot react with alkenes
- [D] Chlorine > Bromine > Iodine

67 [EASY] Addition of Hydrogen Cyanide to Alkenes — 9701/12/M/J/222/Q67**[1]**

Why do alkenes NOT react with hydrogen cyanide (HCN) under normal conditions?

- [A] HCN is an insoluble gas.
- [B] Alkenes have high electron density and repel nucleophilic cyanide ions (:CN⁻); HCN lacks an effective electrophilic species to initiate attack.
- [C] Cyanide ions destroy C-C sigma bonds.
- [D] Alkenes are strong Bronsted bases that neutralize HCN.

68 [HARD] Oxidation of But-2-ene with Hot KMnO₄ — 9701/13/M/J/223/Q68**[1]**

What single organic product is formed when but-2-ene (CH₃-CH=CH-CH₃) is heated with hot concentrated acidified KMnO₄?

- [A] Ethanol, CH₃CH₂OH
- [B] Ethanal, CH₃CHO
- [C] Ethanoic acid, CH₃COOH
- [D] Carbon dioxide and water

69 [EASY] Polyalkene Repeat Unit Deduction — 9701/11/M/J/224/Q69**[1]**

What is the monomer used to produce the polymer with repeat unit -[CH₂-CH(Cl)]-_n?

- [A] 1,2-dichloroethene
- [B] Chloroethene (vinyl chloride), CH₂=CHCl
- [C] Chloroethane
- [D] 1,1-dichloroethene

70 [HARD] Combustion Heat of Alkenes vs Alkanes — 9701/12/M/J/220/Q70**[1]**

Why do alkenes burn with a more smoky (sootier) flame than the corresponding alkanes?

- [A] Alkenes have a higher carbon-to-hydrogen ratio, increasing the likelihood of incomplete combustion to carbon particulates (soot).
- [B] Alkenes contain toxic sulfur impurities.
- [C] Alkenes burn at a lower temperature than alkanes.
- [D] Alkenes produce hydrogen gas during combustion.

71 [EASY] Formation of Diols via OsO₄ vs KMnO₄ — 9701/13/M/J/221/Q71**[1]**

In addition to cold dilute KMnO₄, which reagent can convert an alkene into a vicinal diol via syn-addition?

- [A] Hot concentrated nitric acid
- [B] Acidified potassium dichromate(VI)
- [C] Lithium aluminium hydride
- [D] Osmium tetroxide (OsO₄) followed by reduction

72 [HARD] Asymmetrical Alkene Addition Regioselectivity — 9701/11/M/J/222/Q72**[1]**

In the reaction between 2-methylbut-1-ene and HBr, what percentage of the product is expected to be 2-bromo-2-methylbutane under ionic conditions?

- [A] 50% (equal mixture of 1-bromo and 2-bromo)
- [B] 0% (anti-Markovnikov is favored)
- [C] Virtually 100% (predominant Markovnikov product via the tertiary carbocation)
- [D] 25%

73 [EASY] Anti-Markovnikov Addition Mechanism — 9701/12/M/J/223/Q73**[1]**

Why does the addition of HBr to propene in the presence of peroxides (ROOR) yield 1-bromopropane?

- [A] The reaction switches to a free radical mechanism where the bromine radical (.Br) attacks first, forming the more stable secondary carbon radical.
- [B] Peroxides act as a powerful nucleophile.
- [C] Peroxides reverse the electronegativity of bromine.
- [D] Peroxides destroy the pi bond without adding.

74 [HARD] Addition of Water Across Cyclohexene — 9701/13/M/J/224/Q74**[1]**

What is the product formed when cyclohexene reacts with steam in the presence of concentrated H₃PO₄ catalyst?

- [A] Cyclohexanone
- [B] Hexan-1-ol
- [C] Cyclohexanol
- [D] Adipic acid

75 [EASY] Distinguishing Alkanes from Alkenes — 9701/11/M/J/220/Q75**[1]**

Which single chemical test immediately distinguishes hexane from hex-1-ene?

- [A] Hexane forms a white precipitate with silver nitrate.
- [B] Hex-1-ene decolorizes aqueous bromine water from orange to colorless at room temperature; hexane produces no color change.
- [C] Hex-1-ene reacts with sodium metal to evolve hydrogen.
- [D] Hexane turns blue litmus paper red.

76 [HARD] Oxidative Cleavage of Pent-2-ene — 9701/12/M/J/221/Q76**[1]**

What two organic acids are formed when pent-2-ene, CH₃-CH=CH-CH₂-CH₃, is refluxed with hot concentrated acidified KMnO₄?

- [A] Methanoic acid and butanoic acid
- [B] Two molecules of propanoic acid
- [C] Ethanoic acid and propanone
- [D] Ethanoic acid (CH₃COOH) and propanoic acid (CH₃CH₂COOH)

77 [EASY] Physical State of Small Alkenes — 9701/13/M/J/222/Q77**[1]**

What are the physical states of ethene, propene, and but-1-ene at standard room temperature and pressure (298 K, 100 kPa)?

- [A] Ethene is a gas; propene and but-1-ene are liquids.
- [B] All three are volatile liquids.
- [C] All three are colorless gases.
- [D] Ethene is a liquid; the others are solids.

78 [HARD] Testing for Carbon Dioxide from Cleavage — 9701/11/M/J/223/Q78**[1]**

When an unknown alkene is heated with hot concentrated KMnO₄, the gas evolved turns limewater cloudy. What structural feature does this prove?

- [A] The alkene contains a terminal =CH₂ double-bond group.
- [B] The alkene is a cyclic hydrocarbon.
- [C] The alkene contains a tertiary carbon atom.
- [D] The alkene is completely symmetrical.

79 [EASY] Carbocation Hybridisation and Attack — 9701/12/M/J/224/Q79**[1]**

What is the shape of the carbocation intermediate in electrophilic addition, and from which direction can the nucleophile attack?

- [A] Tetrahedral; attack occurs only from the front.
- [B] Linear; attack occurs along the molecular axis.
- [C] Pyramidal; attack is restricted to the base.
- [D] Trigonal planar; the nucleophile can attack with equal probability from either the top face or bottom face.

80 [HARD] Bromonium Ion vs Open Carbocation — 9701/13/M/J/220/Q80**[1]**

What experimental evidence demonstrates that bromine addition to alkenes involves a cyclic bromonium ion rather than a simple open carbocation?

- [A] The evolution of hydrogen bromide gas.
- [B] The stereospecific formation of trans products (anti-addition) when reacting with cyclic alkenes.
- [C] The formation of equal amounts of cis and trans products.
- [D] The immediate reduction of bromine to bromide.

81 [EASY] Hydration vs Hydrogenation Reaction Conditions — 9701/11/M/J/221/Q81**[1]**

How do the reagents and conditions for converting an alkene to an alcohol differ from those for converting an alkene to an alkane?

- [A] Both use hydrogen gas over a platinum catalyst.
- [B] Alcohol: Steam ($\text{H}_2\text{O}(\text{g})$) with concentrated H_3PO_4 at $300\text{ }^\circ\text{C}$ / 60 atm; Alkane: Hydrogen gas ($\text{H}_2(\text{g})$) with Ni catalyst at $150\text{ }^\circ\text{C}$.
- [C] Both use cold acidified potassium manganate(VII).
- [D] Both use sodium borohydride at room temperature.

82 [HARD] Addition of Halogens in Non-Polar Solvents — 9701/12/M/J/222/Q82**[1]**

When bromine is dissolved in tetrachloromethane (CCl_4) and added to ethene, what is the sole product?

- [A] Bromoethane
- [B] 1-bromo-2-chloroethane
- [C] Ethanol
- [D] 1,2-dibromoethane, $\text{CH}_2\text{Br}-\text{CH}_2\text{Br}$

83 [EASY] Effect of Alkyl Substitution on Alkene Stability — 9701/13/M/J/223/Q83**[1]**

How does increasing alkyl substitution around the $\text{C}=\text{C}$ double bond affect the thermodynamic stability of alkenes?

- [A] Stability increases: tetra-substituted > tri-substituted > di-substituted > mono-substituted > ethene.
- [B] Stability decreases because alkyl groups crowd the double bond.
- [C] All alkene isomers have identical thermodynamic stability.
- [D] Alkyl substitution causes spontaneous decomposition.

84 [HARD] Cleavage Products of 2-methylpent-2-ene — 9701/11/M/J/224/Q84**[1]**

What organic products are formed when 2-methylpent-2-ene, $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_3$, is heated with hot concentrated acidified KMnO_4 ?

- [A] Ethanoic acid and butanone
- [B] Propanal and propanoic acid
- [C] Propanone, $(\text{CH}_3)_2\text{C}=\text{O}$, and propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$
- [D] Carbon dioxide and 2-methylbutanoic acid

85 [EASY] Industrial Polymerisation Conditions — 9701/12/M/J/220/Q85**[1]**

What catalyst and conditions are used in the low-pressure Ziegler-Natta polymerisation of ethene to high-density polyethene (HDPE)?

- [A] Titanium(IV) chloride and triethylaluminium catalyst ($\text{TiCl}_4 / \text{Al}(\text{C}_2\text{H}_5)_3$) at $\sim 60^\circ\text{C}$ and low pressure.
- [B] Concentrated sulfuric acid at 200°C .
- [C] Ultraviolet light in liquid ammonia.
- [D] Iron catalyst at 450°C and 200 atm.

86 [HARD] Electrophilic Addition Transition State — 9701/13/M/J/221/Q86**[1]**

In the reaction coordinate diagram for the electrophilic addition of HBr to an alkene, what represents the highest energy barrier (rate-determining transition state)?

- [A] The attack of the bromide nucleophile on the carbocation.
- [B] The dissolution of HBr in the solvent.
- [C] The diffusion of product molecules away from each other.
- [D] The transition state leading to the formation of the carbocation intermediate.

87 [EASY] Bromination of Phenylethene — 9701/11/M/J/222/Q87**[1]**

When phenylethene (styrene, $\text{C}_6\text{H}_5\text{-CH=CH}_2$) is treated with bromine in the dark at room temperature, what reaction occurs?

- [A] Electrophilic substitution into the benzene ring to form 4-bromostyrene.
- [B] Electrophilic addition across the alkene side chain to form 1,2-dibromo-1-phenylethane; the benzene ring remains unreacted.
- [C] Free radical substitution of the phenyl ring.
- [D] Elimination of hydrogen bromide to form phenylacetylene.

88 [HARD] Hydration of But-1-ene Regiochemistry — 9701/12/M/J/223/Q88**[1]**

What is the major alcohol formed when but-1-ene is reacted with steam in the presence of concentrated phosphoric acid catalyst?

- [A] Butan-1-ol (primary alcohol)
- [B] 2-methylpropan-2-ol
- [C] Butan-2-ol (secondary alcohol)
- [D] Ethanoic acid

89 [EASY] PTFE Monomer and Properties — 9701/13/M/J/224/Q89**[1]**

What is the monomer of poly(tetrafluoroethene) (PTFE / Teflon), and what property makes the polymer non-stick and chemically inert?

- [A] Fluoroethene; weak London dispersion forces.
- [B] Tetrafluoroethene ($\text{CF}_2=\text{CF}_2$); the extremely strong C-F bonds (485 kJ mol^{-1}) and protective fluorine sheath prevent chemical attack.
- [C] 1,1-difluoroethene; high water solubility.
- [D] Trichlorofluoromethane; low boiling point.

90 [HARD] Cleavage of 3-methylhex-3-ene — 9701/11/M/J/220/Q90**[1]**

What organic products are formed when 3-methylhex-3-ene, $\text{CH}_3\text{-CH}_2\text{-C}(\text{CH}_3)=\text{CH-CH}_2\text{-CH}_3$, is refluxed with hot concentrated acidified KMnO_4 ?

- [A] Pentan-3-one ($\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$) and propanoic acid ($\text{CH}_3\text{CH}_2\text{COOH}$)
- [B] Butanone and butanoic acid
- [C] Hexan-3-one and ethanoic acid
- [D] Carbon dioxide and 2-methylpentanoic acid

91 [EASY] Dichromate vs Permanganate Reactivity with Alkenes — 9701/12/M/J/221/Q91**[1]**

Why is acidified potassium manganate(VII) used to test for alkenes rather than acidified potassium dichromate(VI)?

- [A] $K_2Cr_2O_7$ decomposes explosively in the presence of hydrocarbons.
- [B] $KMnO_4$ is a reducing agent that adds hydrogen.
- [C] $K_2Cr_2O_7$ turns colorless upon reduction.
- [D] $KMnO_4$ oxidizes alkenes rapidly at room temperature, producing an immediate color change; $K_2Cr_2O_7$ is a milder oxidizer and does not react readily with C=C bonds at room temperature.

92 [HARD] Markownikoff Addition of HI to Propene — 9701/13/M/J/222/Q92**[1]**

What is the product of the addition of hydrogen iodide (HI) to propene in the dark?

- [A] 1-iodopropane (propyl iodide)
- [B] 1,2-diiodopropane
- [C] 2-iodopropane (isopropyl iodide)
- [D] Cyclopropane and iodine

93 [EASY] Reaction of Ethene with Chlorine Water — 9701/11/M/J/223/Q93**[1]**

When ethene gas is passed into aqueous chlorine water, what is the major organic product formed?

- [A] 2-chloroethanol, CH_2Cl-CH_2OH
- [B] 1,2-dichloroethane
- [C] Chloroethane
- [D] Ethanol

94 [HARD] Catalytic Hydrogenation Stoichiometry of Dienes — 9701/12/M/J/224/Q94**[1]**

What volume of hydrogen gas (measured at RTP) is required to completely hydrogenate 0.50 mol of buta-1,3-diene to butane?

- [A] 12.0 dm³
- [B] 48.0 dm³
- [C] 24.0 dm³
- [D] 6.0 dm³

95 [EASY] Chiral Center Generation in Alkene Addition — 9701/13/M/J/220/Q95**[1]**

Which reaction generates a chiral carbon atom from an achiral starting material?

- [A] Addition of Br_2 to ethene forming 1,2-dibromoethane.
- [B] Addition of HBr to but-1-ene forming 2-bromobutane.
- [C] Addition of H_2 to propene forming propane.
- [D] Addition of HBr to 2-methylpropene forming 2-bromo-2-methylpropane.

96 [HARD] Cleavage of Methylenecyclohexane — 9701/11/M/J/221/Q96**[1]**

What products are obtained when methylenecyclohexane ($=CH_2$ attached to a cyclohexane ring) is oxidized by hot concentrated acidified $KMnO_4$?

- [A] Hexanedioic acid and methanol
- [B] Cyclohexanol and methanoic acid
- [C] Benzoic acid and water
- [D] Cyclohexanone, carbon dioxide, and water

97 [EASY] Combustion Enthalpy of Alkene Isomers — 9701/12/M/J/222/Q97**[1]**

How does the enthalpy of combustion of trans-but-2-ene compare with cis-but-2-ene, and what explains the difference?

- [A] Cis-but-2-ene is significantly more stable because dipoles attract.
- [B] Both have precisely identical combustion enthalpies.
- [C] Trans-but-2-ene is slightly less exothermic (more stable) because the two bulky methyl groups are on opposite sides, minimizing steric repulsion.
- [D] Trans-but-2-ene decomposes spontaneously upon heating.

98 [HARD] Addition of Bromine in the Presence of Sodium Chloride — 9701/13/M/J/223/Q98**[1]**

When ethene reacts with bromine in the presence of concentrated aqueous sodium chloride, which product is formed in addition to 1,2-dibromoethane?

- [A] 1-bromo-2-chloroethane
- [B] 1,2-dichloroethane only
- [C] Bromoethane
- [D] Chloroethane

99 [EASY] Deducing Alkene from Quantitative Cleavage — 9701/11/M/J/224/Q99**[1]**

Complete oxidative cleavage of 0.10 mol of an alkene by hot concentrated KMnO_4 produces 0.10 mol of CO_2 , 0.10 mol of propanone, and 0.10 mol of ethanoic acid. What is the alkene?

- [A] 2-methylbut-2-ene
- [B] Hex-2-ene
- [C] Cyclohexene
- [D] 2-methylpenta-1,4-diene, $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}=\text{CH}_2$ or 4-methylpenta-1,3-diene, $\text{CH}_2=\text{CH}-\text{CH}=\text{C}(\text{CH}_3)_2$

100 [HARD] Industrial Polymerisation of Propene — 9701/12/M/J/220/Q100**[1]**

What is the structure of the repeat unit in poly(propene), and what type of polymerisation reaction forms it?

- [A] $-\text{[CH}_2-\text{CH}_2-\text{CH}_2]-_n$ formed by condensation polymerisation.
- [B] $-\text{[CH}_2-\text{CH}(\text{CH}_3)]-_n$ formed by addition polymerisation.
- [C] $-\text{[CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)]-_n$ formed by free radical substitution.
- [D] $-\text{[C}(\text{CH}_3)_2-\text{CH}_2]-_n$ formed by elimination.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 1: Deducing Alkene Structure from Hot Concentrated KMnO_4 Oxidation Products — 9701/11/M/J/22/Q101

An unsaturated hydrocarbon Y of molecular formula C_7H_{14} is heated with hot, concentrated, acidified potassium manganate(VII). Effervescence is observed, and the gas evolved turns limewater milky. The only organic product remaining in the reaction mixture is 3-methylbutanoic acid, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{COOH}$.

What is the structural formula of hydrocarbon Y?

- [A] 2-methylhex-2-ene, $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3$
- [B] 4-methylhex-1-ene, $\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$
- [C] 5-methylhex-1-ene, $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}(\text{CH}_3)_2$
- [D] 2,4-dimethylpent-2-ene, $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}(\text{CH}_3)_2$

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Regiochemistry and Relative Yields in Unsymmetrical Alkene Addition — 9701/12/M/J/21/Q18 [1]

2-methylbut-2-ene reacts with hydrogen chloride, HCl, in the dark.

Which statement correctly identifies the major product and explains why it forms in higher yield than the minor product?

- [A] 2-chloro-2-methylbutane is the major product because it is formed via a tertiary carbocation intermediate which is more stable than the secondary carbocation intermediate.
- [B] 2-chloro-3-methylbutane is the major product because primary carbocations are more reactive.
- [C] 1-chloro-2-methylbutane is the major product due to steric hindrance at C2.
- [D] An equimolar 50:50 mixture is formed because chlorine attacks both ends of the double bond equally.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Free Radical Chlorination — Relative Abundance of Propagation Intermediates — 9701/13/O/N/22/Q17 [1]

In the photochemical chlorination of propane at 25 °C, hydrogen abstraction can occur at a methyl group or at the central CH₂ group.

- There are 6 methyl hydrogens and 2 methylene hydrogens.
- Secondary C-H bonds are approximately 4 times more reactive towards chlorine radicals than primary C-H bonds.

What is the expected ratio of 1-chloropropane to 2-chloropropane in the product mixture?

- [A] 3 : 1
- [B] 1 : 1
- [C] 1 : 4
- [D] 3 : 4 (or approximately 43% 1-chloropropane and 57% 2-chloropropane)

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Addition of Bromine Water to Polyunsaturated Compounds — 9701/11/F/M/22/Q18 [1]

One mole of an unsaturated hydrocarbon Z reacts completely with exactly two moles of bromine, Br₂, in the dark to form a saturated tetrabromo compound.

Oxidation of one mole of Z with hot concentrated acidified KMnO₄ produces one mole of butanedioic acid (HOOC-CH₂-CH₂-COOH) and two moles of carbon dioxide (CO₂).

What is the structural formula of hydrocarbon Z?

- [A] Hexa-2,4-diene, CH₃-CH=CH-CH=CH-CH₃
- [B] Hexa-1,5-diene, CH₂=CH-CH₂-CH₂-CH=CH₂
- [C] Cyclohexa-1,4-diene
- [D] Hex-1-yne, CH₃-C≡C-CH₂-CH₂-CH₂-CH₃

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Addition of Hydrogen Halide to Alkenes in the Presence of Alcohol Solvents — 9701/12/O/N/21/Q18 [1]

When propene gas is bubbled into a solution of hydrogen bromide dissolved in pure ethanol, what products are formed?

- [A] A mixture of 2-bromopropane, 1-bromopropane, and 2-ethoxypropane
- [B] 2-bromopropane only
- [C] Ethyl bromide and propan-2-ol only
- [D] Propanoic acid and bromoethane

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Stereoisomeric Products of Alkene Addition Reactions — 9701/13/M/J/22/Q18 [1]

When trans-but-2-ene is reacted with hydrogen bromide, HBr, which statement concerning the product is TRUE?

- [A] An optically active single enantiomer of 2-bromobutane is obtained.
- [B] 1-bromobutane is formed as the sole product.
- [C] 2-bromobutane is formed, which contains one chiral center and is produced as a racemic (optically inactive) mixture.
- [D] The product shows cis-trans isomerism.

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Catalytic Cracking Equation and Balancing — 9701/11/O/N/21/Q17**[1]**

One mole of a straight-chain alkane $C_{15}H_{32}$ is cracked to produce one mole of octane (C_8H_{18}), one mole of propene (C_3H_6), and an unknown quantity of ethene (C_2H_4).

How many moles of ethene are produced per mole of $C_{15}H_{32}$?

- [A] 1 mole of ethene
- [B] 2 moles of ethene
- [C] 3 moles of ethene
- [D] 4 moles of ethene

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Distinguishing Alkene Isomers by Oxidation Products — 9701/12/F/M/23/Q18**[1]**

Two isomeric alkenes, P and Q, both have the molecular formula C_5H_{10} .

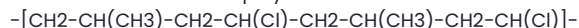
- Alkene P produces a carboxylic acid and a gas that turns limewater milky when heated with hot concentrated acidified $KMnO_4$.
- Alkene Q produces a ketone and a carboxylic acid under the same conditions.

What are the identities of P and Q?

- [A] P is pent-2-ene; Q is 2-methylbut-1-ene
- [B] P is 2-methylbut-1-ene; Q is pent-1-ene
- [C] P is 3-methylbut-1-ene; Q is cyclopentane
- [D] P is pent-1-ene; Q is 2-methylbut-2-ene

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Identifying Addition Polymer Monomers — 9701/13/M/J/23/Q19**[1]**

A section of an addition polymer chain has the structure:



What were the two monomers used in an equimolar ratio to synthesize this co-polymer?

- [A] Propene ($CH_2=CH-CH_3$) and chloroethene ($CH_2=CHCl$)
- [B] Ethene and 2-chloropropane
- [C] But-1-ene and chloroethane
- [D] Prop-2-en-1-ol and chloromethane

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Environmental Role of Catalytic Converters in Hydrocarbon Removal — 9701/11/M/J/23/Q19**[1]**

In a three-way catalytic converter fitted to a car exhaust, unburnt octane (C_8H_{18}) and nitrogen monoxide (NO) react together on the platinum-rhodium surface.

What are the non-toxic products formed by this catalyzed reaction?

- [A] Carbon monoxide (CO), ammonia (NH_3), and oxygen (O_2)
- [B] Carbon soot (C), nitrogen dioxide (NO_2), and steam
- [C] Carbon dioxide (CO_2), water (H_2O), and nitrogen gas (N_2)
- [D] Methane (CH_4), nitric acid (HNO_3), and hydrogen

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. In a limited supply of oxygen, complete oxidation to CO ₂ cannot occur. Instead, incomplete combustion takes place, producing highly toxic carbon monoxide gas (CO, which binds irreversibly to hemoglobin) and unburnt carbon particulates (soot/smoke, causing respiratory distress).	[1]
2	D	Option D is correct. The propagation steps in the chlorination of ethane are: (1) Cl· + CH ₃ CH ₃ → CH ₃ CH ₂ · + HCl; and (2) CH ₃ CH ₂ · + Cl ₂ → CH ₃ CH ₂ Cl + Cl·. Option A is initiation; Option C is termination; Option B incorrectly shows hydrogen radical abstraction (which does not occur because H-Cl bond enthalpy is so much higher than C-Cl).	[1]
3	A	Option A is correct. Catalytic cracking passes vaporised heavy gas oil over a synthetic zeolite (silica/alumina) catalyst at ~500 °C under slight pressure. C-C bonds undergo heterolytic/homolytic cleavage to produce shorter, high-demand fuels (branched alkanes with high octane ratings) and valuable unsaturated chemical feedstocks (alkenes like ethene and propene).	[1]
4	C	Option C is correct. With excess chlorine, successive propagation cycles replace each remaining hydrogen atom on the carbon: CH ₄ → CH ₃ Cl → CH ₂ Cl ₂ → CHCl ₃ → CCl ₄ . With sufficient Cl ₂ and UV irradiation, all four hydrogens are substituted, yielding tetrachloromethane (CCl ₄) as the final major organic product.	[1]
5	A	Option A is correct. The Cl-Cl bond is relatively weak (242 kJ/mol), so UV light of wavelength ~330 nm carries sufficient quantum energy (E=hf) to cause homolytic fission of Cl-Cl, whereas the strong C-H bond (413 kJ/mol) cannot be cleaved directly.	[1]
6	D	Option D is correct. Unburnt volatile organic compounds (VOCs) react with NO _x under sunlight to generate peroxyacetyl nitrate (PAN) and tropospheric ozone, creating irritating photochemical smog.	[1]
7	B	Option B is correct. Thermal cracking operates at severe temperatures (700-1000 °C) and pressures (up to 70 atm) via free radical mechanisms, yielding large fractions of terminal alkenes.	[1]
8	C	Option C is correct. Hydrogen abstraction can form a 1° radical (CH ₃ CH ₂ CH ₂ ·) or a 2° radical (CH ₃ CH·CH ₃). The 2° radical is stabilized by two electron-donating methyl groups (+I effect), lowering the activation energy for its formation.	[1]
9	B	Option B is correct. Equation: C ₃ H ₈ (g) + 5O ₂ (g) → 3CO ₂ (g) + 4H ₂ O(l). By Avogadro's law, gas volume ratios equal mole ratios at the same T and P. Thus, 1.0 dm ³ of C ₃ H ₈ requires 5.0 dm ³ of O ₂ .	[1]
10	A	Option A is correct. Methane molecules possess C-H stretching and bending vibrational modes that absorb infrared radiation strongly in the thermal infrared window, trapping re-radiated heat.	[1]
11	D	Option D is correct. Crude oil is separated by fractional distillation: smaller hydrocarbons have fewer electrons, weaker London dispersion forces, and lower boiling points, condensing near the top of the tower.	[1]
12	C	Option C is correct. The C-C and C-H bonds have high bond energies (348 and 413 kJ/mol) and almost identical electronegativities (C=2.5, H=2.1), leaving no polar delta+ sites or lone pairs for reagents to attack.	[1]
13	A	Option A is correct. Straight-chain alkanes knock easily in car engines. Catalytic reforming (using Pt catalyst at 500 °C) isomerises them into branched and aromatic rings (like benzene/toluene), greatly boosting octane numbers.	[1]
14	C	Option C is correct. Combustion equation: C ₅ H ₁₂ + 8O ₂ → 5CO ₂ + 6H ₂ O. Each mole of C ₅ H ₁₂ contains 12 hydrogen atoms, producing 6 moles of H ₂ O.	[1]
15	B	Option B is correct. The sum of the two propagation steps equals the overall reaction (CH ₄ + Cl ₂ → CH ₃ Cl + HCl), which is exothermic (delta-H ~ -100 kJ/mol), sustaining the reaction without further initiation once radicals exist.	[1]
16	D	Option D is correct. With a large excess of methane, a chlorine radical is far more likely to collide with an unreacted CH ₄ molecule than with a CH ₃ Cl molecule, minimizing poly-substitution.	[1]
17	C	Option C is correct. Water molecules form extensive hydrogen bond networks. Hexane cannot form hydrogen bonds to disrupt water-water attractions, but easily dissolves non-polar compounds via London dispersion forces.	[1]
18	A	Option A is correct. Monochlorination of propane yields two positional isomers: 1-chloropropane (substitution at C1/C3) and 2-chloropropane (substitution at C2).	[1]
19	D	Option D is correct. Termination involves radical pairing (R· + Br· → RBr, or R· + R· → R-R). Free hydrogen atoms (·H) are never formed in organic free radical mechanisms, so H ₂ is never formed as a termination product.	[1]
20	B	Option B is correct. Fluorination is violently explosive even in the dark (delta-H highly exothermic, low activation energy); iodination is endothermic and does not take place because I· is too unreactive to abstract hydrogen.	[1]
21	B	Option B is correct. Branched alkanes produce more stable intermediate radicals and burn with a controlled flame front, resisting auto-ignition (knocking) under engine compression.	[1]
22	D	Option D is correct. Methane has the highest H:C ratio (4:1) of any hydrocarbon. A higher proportion of its combustion heat comes from forming O-H bonds (in H ₂ O) rather than C=O bonds (in CO ₂), emitting less CO ₂ per MJ of energy.	[1]
23	A	Option A is correct. As chain length grows, polarisability increases, London dispersion forces strengthen, and long chains become physically entangled, causing viscosity to increase steadily.	[1]
24	C	Option C is correct. Molecular oxygen (O ₂) has diradical character and scavenges alkyl radicals (·CH ₃ + O ₂ → CH ₃ OO·), which are far less reactive and cannot propagate the chain until oxygen is consumed.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
25	A	Option A is correct. Conservation of atoms: $C_{12}H_{26} \rightarrow C_8H_{18} + 2(C_2H_4)$. Carbons: $12 - 8 = 4$, divided by $2 = 2$ (C_2H_4 , ethene).	[1]
26	D	Option D is correct. C_6H_{12} requires 6 moles of O_2 for carbon (forming $6CO_2$) and 3 moles of O_2 for hydrogen (forming $6H_2O$). Total $O_2 = 6 + 3 = 9$ moles.	[1]
27	B	Option B is correct. Ozone is a secondary pollutant formed in the troposphere via photochemical reactions of NO_2 and VOCs in sunlight; it is not emitted directly from car exhausts.	[1]
28	C	Option C is correct. Alkyl groups donate electron density toward the electron-deficient trivalent carbon atom via inductive effect and hyperconjugation, stabilizing $3^\circ > 2^\circ > 1^\circ$ radicals.	[1]
29	B	Option B is correct. Branching makes molecules more compact and spherical. Spheres touch at only one point, reducing molecular surface contact area. Less surface contact produces weaker London dispersion forces, lowering the boiling point.	[1]
30	A	Option A is correct. Each additional $-CH_2-$ group requires 1.5 extra O_2 molecules and forms one more CO_2 and H_2O , adding approximately -650 kJ/mol of exothermic energy.	[1]
31	D	Option D is correct. Coupling of two $(CH_3)_3C\cdot$ radicals forms $(CH_3)_3C-C(CH_3)_3$, which is 2,2,3,3-tetramethylbutane.	[1]
32	C	Option C is correct. Particulate matter ≤ 2.5 micrometers ($PM_{2.5}$) bypasses nasal filtration, penetrating deep into lung alveoli and causing severe cardiovascular and respiratory diseases and cancer.	[1]
33	A	Option A is correct. Alkanes are saturated and lack electron-rich pi bonds; they are completely resistant to oxidation by cold acidified $KMnO_4$.	[1]
34	C	Option C is correct. Petroleum (crude oil) and natural gas are the overwhelming industrial sources of alkanes, formed by geothermal heating and compression of prehistoric marine biomass over millions of years.	[1]
35	B	Option B is correct. By Hammond's postulate, because hydrogen abstraction by Br $\cdot$ is endothermic, the transition state resembles the radical intermediate, making the energy difference between 1° and $2^\circ/3^\circ$ pathways much larger (high regioselectivity).	[1]
36	D	Option D is correct. The oxidation catalyst (Pt/Pd) oxidizes unburnt hydrocarbons and toxic CO into harmless CO_2 and H_2O : $C_xH_y + O_2 \rightarrow CO_2 + H_2O$; $2CO + O_2 \rightarrow 2CO_2$.	[1]
37	C	Option C is correct. Methane clathrate is a solid lattice of water molecules physically encaging methane gas molecules, stable under high hydrostatic ocean pressures and near-freezing temperatures.	[1]
38	A	Option A is correct. While ΔG for alkane combustion is large and negative, the reaction does not occur at $25^\circ C$ without a spark because the activation energy (breaking strong covalent bonds) is very high, imparting kinetic stability.	[1]
39	D	Option D is correct. Without UV light (or temperatures $> 300^\circ C$), Cl-Cl bonds cannot undergo homolytic cleavage to generate the initiating chlorine radicals, so the mixture remains completely unreacted.	[1]
40	B	Option B is correct. Alkenes contain reactive $C=C$ double bonds, making them prime feedstocks for synthesis of plastics, synthetic rubbers, epoxy resins, antifreeze, and pharmaceuticals.	[1]
41	B	Option B is correct. The high electron density of the $C=C$ pi bond induces a dipole in the Br_2 molecule ($Br(\delta^+) - Br(\delta^-)$). The δ^+ bromine acts as an electrophile, accepting electrons from the pi bond to form a carbocation / bromonium ion intermediate, which is attacked by Br^- to yield 1,2-dibromoethane. The orange color of bromine disappears (decolorization).	[1]
42	D	Option D is correct. Electrophilic addition of H^+ can form either a primary carbocation ($CH_3CH_2CH_2^+$) or a secondary carbocation ($CH_3CH^+CH_3$). The secondary carbocation has two electron-donating methyl groups (+I inductive effect) that disperse the positive charge, lowering its activation energy. Attack by Br^- on the 2° carbocation yields 2-bromopropane as the major product.	[1]
43	A	Option A is correct. Cold, dilute $KMnO_4$ acts as a mild oxidizing agent that hydrolyzes the alkene across the double bond without cleaving the $C-C$ sigma bond, adding two $-OH$ groups to give a vicinal diol (propane-1,2-diol). The purple MnO_4^- ion is reduced to colorless Mn^{2+} (in acid) or brown MnO_2 precipitate (in neutral/alkaline medium).	[1]
44	C	Option C is correct. Under hot concentrated $KMnO_4$, the $C=C$ bond is completely cleaved: $=CR_2$ fragments oxidize to ketones ($R_2C=O$), $=CHR$ fragments oxidize to carboxylic acids ($RCOOH$), and $=CH_2$ fragments oxidize to $CO_2 + H_2O$. Since the products are propanone ($(CH_3)_2C=O$) and ethanoic acid (CH_3COOH), the original alkene must be $(CH_3)_2C=CH-CH_3$ (2-methylbut-2-ene).	[1]
45	A	Option A is correct. The industrial hydration of ethene ($C_2H_4 + H_2O \rightarrow C_2H_5OH$) uses steam over a solid H_3PO_4 catalyst supported on silica at $300^\circ C$ and 60 atm.	[1]
46	D	Option D is correct. Hydrogenation of liquid vegetable oils (containing polyunsaturated alkenes) over a nickel catalyst at $\sim 150^\circ C$ saturates $C=C$ bonds, raising the melting point to produce semi-solid margarine.	[1]
47	B	Option B is correct. In aqueous bromine, water molecules are present in overwhelming concentration compared to Br^- ions. After the electrophile Br^+ attacks to form the secondary carbocation ($CH_3-CH^+-CH_2Br$), nucleophilic H_2O attacks the 2° carbon, forming a halohydrin: 1-bromopropan-2-ol.	[1]
48	C	Option C is correct. Under hot concentrated $KMnO_4$, a terminal $=CH_2$ group is initially oxidized to methanoic acid ($HCOOH$), which undergoes rapid further oxidation by $KMnO_4$ into carbon dioxide (CO_2) and water (H_2O), producing visible effervescence.	[1]
49	B	Option B is correct. The symmetrical alkene $(CH_3)_2C=C(CH_3)_2$ cleaves across the $C=C$ bond. Both sides consist of $=C(CH_3)_2$ groups, yielding two molecules of propanone ($(CH_3)_2C=O$).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
50	A	Option A is correct. Addition of Cl ₂ across the C ₂ =C ₃ double bond adds one chlorine to each carbon, yielding 2,3-dichlorobutane.	[1]
51	D	Option D is correct. Addition of H ⁺ forms (CH ₃) ₃ C-CH ⁺ -CH ₃ (a 2° carbocation). A methyl group migrates with its electron pair (1,2-hydride/methyl shift) to produce (CH ₃) ₂ C ⁺ -CH(CH ₃) ₂ , which is tertiary and much more stable.	[1]
52	C	Option C is correct. Bromine water tests for C=C unsaturation: addition across the double bond rapidly decolorizes the orange-brown bromine solution at room temperature without requiring UV light or catalysts.	[1]
53	A	Option A is correct. Alkyl groups are electron-donating (+I). The 3° carbocation has three alkyl groups dispersing the positive charge, making it far more stable than the 1° carbocation with only one alkyl group.	[1]
54	C	Option C is correct. Chlorine is more electronegative than iodine (Cl=3.0, I=2.5), so I carries a partial positive charge (I delta+ - Cl delta-). The pi bond attacks I ⁺ , forming the more stable secondary carbocation CH ₃ -CH ⁺ -CH ₂ I, which is then attacked by Cl ⁻ to yield 2-chloro-1-iodopropane.	[1]
55	B	Option B is correct. A single product of butanoic acid indicates a symmetrical alkene where both halves are =CH-CH ₂ CH ₂ CH ₃ . The original alkene is oct-4-ene.	[1]
56	D	Option D is correct. Addition polymerisation occurs when thousands of alkene monomers join together: the pi bond in each C=C breaks to form new C-C sigma bonds linking the monomers into a long saturated polymer chain.	[1]
57	C	Option C is correct. Electrophilic attack by Br ₂ forms a three-membered cyclic bromonium ion that blocks one face of the ring. The bromide nucleophile must attack from the opposite face (backside attack), resulting strictly in anti-addition (trans product).	[1]
58	A	Option A is correct. Ethene reacts with concentrated H ₂ SO ₄ via electrophilic addition to form ethyl hydrogensulfate (CH ₃ CH ₂ OSO ₃ H). Adding water and warming hydrolyzes the ester to ethanol and regenerates sulfuric acid: CH ₃ CH ₂ OSO ₃ H + H ₂ O → CH ₃ CH ₂ OH + H ₂ SO ₄ .	[1]
59	D	Option D is correct. Benzene possesses ~150 kJ/mol of aromatic resonance stabilization energy. Electrophilic addition would break the delocalized ring, permanently destroying aromaticity. Hence, benzene undergoes substitution to regenerate the stable 6 pi aromatic ring.	[1]
60	B	Option B is correct. Cleaving the C=C bond in cyclohexene breaks open the ring. Both double-bond carbons are =CH- groups, so each is oxidized to a carboxylic acid group (-COOH), yielding the open-chain dicarboxylic acid hexanedioic acid.	[1]
61	B	Option B is correct. As Br ₂ nears the electron-rich pi cloud, electrostatic repulsion pushes electrons away from the closer Br atom to the further Br atom, creating an induced dipole (Br delta+ - Br delta-).	[1]
62	D	Option D is correct. Protonation of (CH ₃) ₂ C=CH ₂ forms the tertiary carbocation (CH ₃) ₃ C ⁺ (much more stable than the 1° carbocation). Attack by H ₂ O yields 2-methylpropan-2-ol.	[1]
63	A	Option A is correct. The terminal =CH ₂ fragment oxidizes to CO ₂ + H ₂ O. The =C(CH ₃)CH ₂ CH ₃ fragment oxidizes to the ketone butanone (CH ₃ COCH ₂ CH ₃).	[1]
64	C	Option C is correct. The rate-determining step involves breaking the H-X bond as H ⁺ is transferred to the alkene pi bond. Because H-Cl has a stronger bond than H-Br, more energy is required, making the reaction slower.	[1]
65	A	Option A is correct. Equation for alkene C _n H _{2n} : C _n H _{2n} + 1.5n O ₂ → n CO ₂ + n H ₂ O. Volume ratio O ₂ / alkene = 1.5n / 20 = 4.5. Thus, n = 4.5 / 1.5 = 3 (propene, C ₃ H ₆).	[1]
66	D	Option D is correct. Electrophilic reactivity depends on the electrophilicity and polarizability of the halogen; chlorine reacts more rapidly than bromine, which reacts faster than iodine.	[1]
67	B	Option B is correct. Alkenes are electron-rich nucleophiles and require an initial electrophilic attack. HCN contains a weak, poorly electrophilic H ⁺ and a nucleophilic :CN ⁻ ion, so no reaction occurs (unlike carbonyls which are electrophilic).	[1]
68	C	Option C is correct. Both halves of the symmetrical but-2-ene molecule consist of =CH-CH ₃ groups, which are oxidized to carboxylic acid groups, producing two molecules of ethanoic acid (CH ₃ COOH).	[1]
69	B	Option B is correct. Restoring the C=C double bond between the two carbon atoms of the repeat unit gives CH ₂ =CHCl (chloroethene).	[1]
70	A	Option A is correct. Ethene (C:H = 1:2) has a higher proportion of carbon than ethane (C:H = 1:3). More oxygen is required per gram to achieve complete oxidation, resulting in a luminous, smoky flame due to incandescent soot particles.	[1]
71	D	Option D is correct. Osmium tetroxide (OsO ₄) selectively oxidizes alkenes to 1,2-diols via a cyclic osmate ester, giving stereospecific syn-dihydroxylation.	[1]
72	C	Option C is correct. The difference in activation energy between forming the 3° carbocation and the 1° carbocation is so large (~40 kJ/mol) that the 3° pathway overwhelmingly dominates, producing almost exclusively the tertiary haloalkane.	[1]
73	A	Option A is correct. In the presence of peroxides, RO· radicals generate ·Br. The ·Br radical adds to C1 of propene to produce the more stable 2° radical (CH ₃ -CH·-CH ₂ Br), which abstracts H from HBr to form 1-bromopropane (anti-Markovnikov).	[1]
74	C	Option C is correct. Electrophilic addition of H ₂ O across the C=C double bond of cyclohexene yields the cyclic alcohol cyclohexanol.	[1]
75	B	Option B is correct. Hex-1-ene contains an unsaturated C=C bond that rapidly undergoes electrophilic addition with aqueous bromine, decolorizing it. Saturated hexane shows no reaction in the absence of UV light.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
76	D	Option D is correct. Cleavage of the C2=C3 bond divides the molecule into a 2-carbon =CH-CH3 fragment (oxidized to ethanoic acid) and a 3-carbon =CH-CH2CH3 fragment (oxidized to propanoic acid).	[1]
77	C	Option C is correct. The C2 to C4 alkenes (ethene, propene, but-1-ene, but-2-ene, 2-methylpropene) have low molecular masses and weak London dispersion forces, existing as gases at RTP.	[1]
78	A	Option A is correct. Only terminal =CH2 groups are oxidized to CO2 gas (and H2O) by hot concentrated KMnO4. The evolution of CO2 (which precipitates CaCO3 in limewater) proves the presence of a terminal =CH2 group.	[1]
79	D	Option D is correct. The positively charged carbon of a carbocation is sp2 hybridised with an empty unhybridised p orbital perpendicular to the trigonal plane (120°). Nucleophiles can attack this empty p lobe from either the top or bottom with equal probability.	[1]
80	B	Option B is correct. If an open, freely rotating carbocation formed, nucleophilic attack would occur from both faces to give a mixture of cis and trans isomers. The exclusive formation of trans products proves the presence of a bridged bromonium ion preventing rotation and blocking syn attack.	[1]
81	B	Option B is correct. Alkene → alcohol is hydration (addition of steam over solid H3PO4 catalyst at high temperature and pressure); alkene → alkane is hydrogenation (addition of H2 gas over Ni catalyst at ~150 °C).	[1]
82	D	Option D is correct. In an inert non-polar solvent like CCl4, no competing nucleophiles (like water) exist. The only nucleophile available to attack the bromonium ion is the bromide ion (Br-), forming 1,2-dibromoethane as the exclusive product.	[1]
83	A	Option A is correct. Alkyl groups donate electron density to the pi system via hyperconjugation and strengthen bonding, making more highly substituted alkenes (like (CH3)2C=C(CH3)2) thermodynamically more stable than less substituted isomers.	[1]
84	C	Option C is correct. Cleaving the double bond in (CH3)2C=CH-CH2-CH3 produces propanone from the (CH3)2C= fragment and propanoic acid from the =CH-CH2-CH3 fragment.	[1]
85	A	Option A is correct. Ziegler-Natta coordination catalysts (TiCl4 with Al(C2H5)3) enable coordination polymerisation of ethene at modest temperatures (~60 °C) and atmospheric pressure, producing unbranched linear HDPE with high density and tensile strength.	[1]
86	D	Option D is correct. The rate-determining step is the initial protonation of the alkene pi bond to form the high-energy carbocation intermediate. This step has the highest activation energy on the reaction coordinate.	[1]
87	B	Option B is correct. The aliphatic alkene double bond has high, localized electron density and reacts rapidly with bromine via electrophilic addition, whereas the aromatic benzene ring requires a halogen carrier catalyst (AlBr3) and does not react in the dark.	[1]
88	C	Option C is correct. Addition of H+ to but-1-ene forms the secondary carbocation CH3-CH2-CH+-CH3 in preference to the primary carbocation CH3-CH2-CH2-CH2+ due to stabilization by two alkyl groups. Attack by H2O gives butan-2-ol as the major product.	[1]
89	B	Option B is correct. PTFE is synthesized from CF2=CF2. The C-F bond is one of the strongest single bonds in organic chemistry (485 kJ/mol), and the small, highly electronegative fluorine atoms tightly pack around the carbon backbone, forming an unreactive, non-stick barrier.	[1]
90	A	Option A is correct. Cleaving across C3=C4 yields pentan-3-one from the =C(CH3)CH2CH3 carbon and propanoic acid from the =CH-CH2CH3 carbon.	[1]
91	D	Option D is correct. Potassium manganate(VII) is a much stronger oxidizing agent than potassium dichromate(VI) (E° +1.51 V vs +1.33 V) and readily attacks alkene pi bonds at room temperature, while K2Cr2O7 reacts sluggishly or not at all with isolated C=C bonds.	[1]
92	C	Option C is correct. HI adds via electrophilic addition. Protonation yields the secondary carbocation CH3-CH+-CH3, which is captured by iodide to form 2-iodopropane as the dominant Markovnikov product.	[1]
93	A	Option A is correct. In aqueous chlorine, the electrophile Cl+ attacks ethene to form the chloronium ion intermediate +CH2-CH2Cl. Because water molecules are in overwhelming excess compared to chloride ions, H2O attacks the intermediate, forming 2-chloroethanol (a chlorohydrin).	[1]
94	C	Option C is correct. Buta-1,3-diene contains two C=C double bonds. Complete saturation requires 2 moles of H2 per mole of diene: 0.50 mol × 2 = 1.0 mol of H2. At RTP, 1.0 mol of any gas occupies 24.0 dm³.	[1]
95	B	Option B is correct. But-1-ene is achiral. Addition of HBr yields 2-bromobutane (CH3-CH(Br)-CH2CH3). C2 is bonded to -H, -Br, -CH3, and -CH2CH3 (four different groups), creating a chiral center.	[1]
96	D	Option D is correct. The exocyclic =CH2 group oxidizes to CO2 + H2O. The ring carbon of the double bond (=C<) is bonded to two ring carbons, oxidizing to the cyclic ketone cyclohexanone.	[1]
97	C	Option C is correct. In cis-but-2-ene, the two methyl groups are on the same side of the double bond, causing mutual steric hindrance (van der Waals repulsion). In trans-but-2-ene, the methyl groups are opposite (180°), relieving steric strain. Trans is ~4 kJ/mol more stable, releasing less heat upon combustion.	[1]
98	A	Option A is correct. The electrophile Br+ attacks ethene to form the bromonium ion. In the second step, either Br-, Cl- (from NaCl), or H2O can act as the nucleophile. Attack by Cl- yields 1-bromo-2-chloroethane.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
99	D	Option D is correct. Cleavage of 4-methylpenta-1,3-diene ($\text{CH}_2=\text{CH}-\text{CH}=\text{C}(\text{CH}_3)_2$) produces: CO_2 from terminal $=\text{CH}_2$; ethanoic acid from the middle $=\text{CH}-\text{CH}=$ segment (or two segments); and propanone from $=\text{C}(\text{CH}_3)_2$. This matches the 1:1:1 mole ratio of products.	[1]
100	B	Option B is correct. Propene ($\text{CH}_2=\text{CH}-\text{CH}_3$) undergoes addition polymerisation: the $\text{C}=\text{C}$ pi bond breaks and monomers link through $\text{C}-\text{C}$ sigma bonds to form poly(propene), having repeat unit $-\text{CH}_2-\text{CH}(\text{CH}_3)-$.	[1]
101	C	Option C is correct. The evolution of CO_2 (which turns limewater milky) proves that hydrocarbon Y contains a terminal $=\text{CH}_2$ group. The only organic product is 3-methylbutanoic acid, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{COOH}$ (5 carbons). Reconnecting the $=\text{CH}_2$ carbon to the carbonyl carbon of the acid yields: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}(\text{CH}_3)_2$ (5-methylhex-1-ene, total 7 carbons).	[1]
102	A	Option A is correct. Addition of H^+ to 2-methylbut-2-ene ($(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_3$) can occur at C_3 to form the tertiary carbocation $(\text{CH}_3)_2\text{C}^+-\text{CH}_2\text{CH}_3$, or at C_2 to form the secondary carbocation $(\text{CH}_3)_2\text{CH}-\text{CH}^+-\text{CH}_3$. The 3° carbocation is stabilized by three alkyl groups (+I effect), lowering the activation energy barrier. Attack by Cl^- gives 2-chloro-2-methylbutane as the predominant product.	[1]
103	D	Option D is correct. Relative rate = (number of hydrogens) * (relative reactivity). For 1-chloropropane: 6 primary $\text{H} \times 1 = 6$. For 2-chloropropane: 2 secondary $\text{H} \times 4 = 8$. Ratio of 1-chloropropane to 2-chloropropane = $6 : 8 = 3 : 4$ ($3/7 = 42.9\%$ vs $4/7 = 57.1\%$).	[1]
104	B	Option B is correct. Reacting with 2 moles of Br_2 to form a tetrabromo alkane confirms 2 degrees of unsaturation (two $\text{C}=\text{C}$ double bonds). The products of hot KMnO_4 oxidation are two moles of CO_2 (proving TWO terminal $=\text{CH}_2$ groups) and one mole of $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{COOH}$ (4 carbons in the central chain). Reassembling the fragments: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}=\text{CH}_2$ (hexa-1,5-diene).	[1]
105	A	Option A is correct. H^+ from HBr adds to propene to form the carbocations $\text{CH}_3-\text{CH}^+-\text{CH}_3$ (major) and $\text{CH}_3-\text{CH}_2-\text{CH}_2^+$ (minor). The carbocations can be attacked by bromide ions (yielding 2-bromopropane and 1-bromopropane) OR by the nucleophilic solvent ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), which upon deprotonation yields the ether 2-ethoxypropane.	[1]
106	C	Option C is correct. Addition of HBr to but-2-ene gives 2-bromobutane, $\text{CH}_3-\text{CH}(\text{Br})-\text{CH}_2-\text{CH}_3$. C_2 is a chiral center bonded to $-\text{H}$, $-\text{Br}$, $-\text{CH}_3$, and $-\text{CH}_2\text{CH}_3$. Because the intermediate carbocation is planar (sp^2), the bromide ion can attack from either face with equal probability (50% top, 50% bottom), yielding an equimolar racemic mixture that has zero net optical rotation.	[1]
107	B	Option B is correct. Equation: $\text{C}_{15}\text{H}_{32} \rightarrow \text{C}_8\text{H}_{18} + \text{C}_3\text{H}_6 + n \text{C}_2\text{H}_4$. Balancing carbon atoms: $15 = 8 + 3 + 2n \Rightarrow 15 - 11 = 4 \Rightarrow 2n = 4 \Rightarrow n = 2$. Balancing hydrogen atoms: $32 = 18 + 6 + 2(4) = 32$ (verified). Thus, 2 moles of ethene are produced.	[1]
108	D	Option D is correct. Alkene P produces CO_2 gas (from a terminal $=\text{CH}_2$) and a carboxylic acid (butanoic acid), identifying P as pent-1-ene ($\text{CH}_2=\text{CH}-\text{CH}_2\text{CH}_2\text{CH}_3$). Alkene Q produces a ketone and a carboxylic acid, requiring a $=\text{C}(\text{R})_2$ group and a $=\text{CHR}$ group; 2-methylbut-2-ene ($(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_3$) cleaves to propanone (ketone) and ethanoic acid (carboxylic acid).	[1]
109	A	Option A is correct. Dividing the polymer chain into two-carbon repeating segments: the first repeat unit is $-\text{CH}_2-\text{CH}(\text{CH}_3)-$ (which derives from the monomer propene, $\text{CH}_2=\text{CH}-\text{CH}_3$) and the second repeat unit is $-\text{CH}_2-\text{CH}(\text{Cl})-$ (which derives from chloroethene / vinyl chloride, $\text{CH}_2=\text{CHCl}$). The alternating structure indicates co-polymerisation of propene and chloroethene.	[1]
110	C	Option C is correct. The catalytic converter simultaneously oxidizes unburnt hydrocarbons and reduces nitrogen oxides: $\text{C}_8\text{H}_{18} + 25\text{NO} \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O} + 12.5\text{N}_2$. All three resulting products (CO_2 , H_2O , and N_2) are non-toxic atmospheric constituents.	[1]